Mandar Ambulkar

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EDUCATION

Northeastern University, San Jose, CA - Master of Science in Computer Science, GPA: 3.8/4.0 **Jan 2024 – Dec 2025**
Relevant Courses: *Pattern Recognition, Computer Vision, NLP, Data Mining, Algorithms, Programming Design Paradigm.*

Vishwakarma Institute of Technology, Pune, India **Aug 2019 – May 2023**
Bachelor of Technology, Information Technology, GPA: 8.47/10.0
Relevant Coursework: *Database Management Systems, Machine Learning, Deep Learning, Cloud Computing, Web Development, Networking.*

EXPERIENCE

Software Engineer, Tata Consultancy Services (Client: Stellantis) | Auburn Hills, MI ***Sep 2026 – Present***

- **Spearheaded onboarding of Pinewood AI DMS** into Stellantis' aftersales dealer data integration platform, ingesting 10+ feeds (repair orders, parts, labor) via Python/AWS Glue ETL jobs mapped to the target schema in S3.
- **Reduced malformed-record ingestion by ~90%** by building a Python schema-validation layer with SQS dead-letter routing, ensuring only conformant data reaches the downstream critical service management application.
- **Cut integration test turnaround by 40%** by building Postman collections and Dockerized Node.js mock services replicating Pinewood API responses, enabling repeatable contract testing of DMS endpoints across dev/QA environments.
- **Resolved data-mapping discrepancies within 48 hours** by coordinating Stellantis SMEs and offshore TCS teams, owning the source-to-target mapping spec, and reporting schema changes and risks to managers weekly.

Software Development Engineer, PASAL Inc. | Remote ***May 2026 – August 2026***

- **Architected and deployed** a scalable, distributed, cloud-native AI outbound email pipeline using Next.js 15, Inngest durable functions, and LLMs (Gemini & GPT-4o-mini), automating ~500 personalized emails/month at ~$85/month.
- **Engineered a robust database layer** leveraging Neon Postgres and Drizzle ORM, integrating a non-bypassable pre-send compliance verification framework to ensure 100% adherence to CAN-SPAM and AI-content regulations.

Software Development Engineer, SurfSense (NU ChatBot Project) | San Jose ***September 2025 – December 2025***

- **Productionized an open-source RAG platform** with RBAC, domain-restricted authentication, and privacy-aware retrieval on Azure (App Services, PostgreSQL, AD), enabling secure, compliant multi-tenant access for 1,000+ institutional users.
- **Reduced end-to-end query latency** from 320 ms to <180 ms by optimizing a hybrid RAG pipeline using pgvector, rerankers, intelligent response caching, and async FastAPI endpoints under concurrent production load.
- **Improved frontend performance by 40%** by designing a **modular** React + Zustand architecture with multi-session chat memory visualization for scalable context handling across thousands of interactions.

Graduate Teaching Assistant, Khoury College of Computer Science | Northeastern University **September 2024 – April 2025**

- Led a team of 3 TAs to support 40+ students in algorithm design and analysis (CS5800/CS5008) via labs, office hours, and programming guidance, co-developing course materials with faculty.

Software Developer, Reticulo Sports Technology Pvt. Ltd. | Pune, India **January 2023- December 2023**

- **Improved product categorization efficiency by 25%** by classifying 12,000+ images into 12 categories with MobileNetV3, cutting manual verification time by 30%.
- **Boosted GPU inference speed by 40%** using Nvidia Triton and CUDA, reducing latency from 15ms to 10ms while enhancing throughput by 20%.
- **Deployed optimized deep learning models** on low-power mobile devices, reducing memory usage by 50% and halving model size (1.2GB → 600MB).

PROJECTS

Interview Assistant – AI Hiring Platform [Link] **Personal Project | 2026**
Two-sided platform: employer assessment studio and proctored candidate mock-interview workspace

- **Delivered real-time interview feedback under 2 s end-to-end** by streaming Whisper speech-to-text and locally hosted Ollama LLM evaluations over WebSockets and SSE to a React 18 + TypeScript client.
- **Built a full-stack hiring platform with role-based employer and candidate workflows** using FastAPI REST APIs and Tailwind CSS, covering assessment authoring, proctored sessions, and automated scoring.

AI Demo Builder (Cloud Computing Project) [Link] **San Jose | December 2025**
Cloud-native, serverless AI platform for automated demo generation from GitHub repositories

- Architected a distributed serverless platform using 18 event-driven AWS Lambda microservices with S3 and DynamoDB, provisioned end-to-end via AWS CDK for reproducible, production-grade infrastructure.

ACHIEVEMENTS / PUBLICATIONS

- **3rd place, Northeastern Student Research Showcase, 2024: Multi Modal Hand Gesture Recognition** [link]
  - Achieved 97.25% validation accuracy by developing ensemble of 34 binary classifiers with priority-based fusion, processing 119GB dataset with 25-30 FPS real-time performance and <40ms latency.
- **Publication: Number Plate Recognition System, IC-ICN 2022 [link]**
- **Publication: KhetiKaksha (Farmer Hub), IC-ICN 2022 [link]**

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, C
Web & Backend: Angular, Node.js, React, Next.js, FastAPI, Flask, REST APIs, Microservices, OAuth, WebSockets, Tailwind CSS, Jest
Cloud & Data Engineering: AWS (S3, Glue, SQS, Lambda, CloudWatch, CDK), Azure, GCP, ETL Pipelines, Data Integration, Data Validation, PySpark
Databases: PostgreSQL, MySQL, MongoDB, Snowflake, DynamoDB, SQLite, Supabase, pgvector
DevOps & Tools: Docker, Kubernetes, Terraform, Jenkins, GitLab CI, CI/CD, Git, Postman, Prometheus, Grafana, Agile/Scrum
AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LLM Integration, RAG, FAISS, NVIDIA Triton, TensorRT, OpenVINO, Pandas, NumPy